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TensorFlow ResNet

Building, Training and Scaling Residual Networks on TensorFlow

TensorFlow

TensorFlow ResNet: Building, Training and Scaling Residual Networks on TensorFlow

ResNet won first place in the Large Scale Visual Recognition Challenge (ILSVRC) in 2015. It was the first neural network not affected by the “vanishing gradient” problem. TensorFlow makes it easy to build ResNet models: you can run pre-trained ResNet-50 models, or build your own custom ResNet implementation. We’ll show you how to do this with ImageNet or CIFAR-10 datasets, and how to automate the process through MissingLink’s deep learning platform.

In this article you will learn

What is ResNet?

Understand Identity Shortcut Connections

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What is ResNet?

Residual Network (ResNet) is a Convolutional Neural Network (CNN) architecture, designed to train very deep neural networks. In theory, a deeper neural network should perform better on the training set because of additional layers processing smaller and smaller features. In reality, the frequency of training errors increases the network is too deep. With ResNet, the training error actually decreases as the network gets deeper.

ResNet provides a breakthrough solution to the “vanishing gradient” problem. Vanishing gradient is a difficulty encountered when you train artificial neural networks with gradient-based methods like backpropagation. With these methods, the gradients of the loss function approach zero as you add more layers to the network. This makes it hard to learn and tune the parameters of the earlier layers in the network.

What are Identity Shortcut Connections?

ResNet is based on “shortcut connections”. This is a way to skip the training of one or more layers—creating a residual block.

Residual blocks allow you to train much deeper neural networks. ResNet structured by taking many of these blocks and stacking them together to form a deep network.

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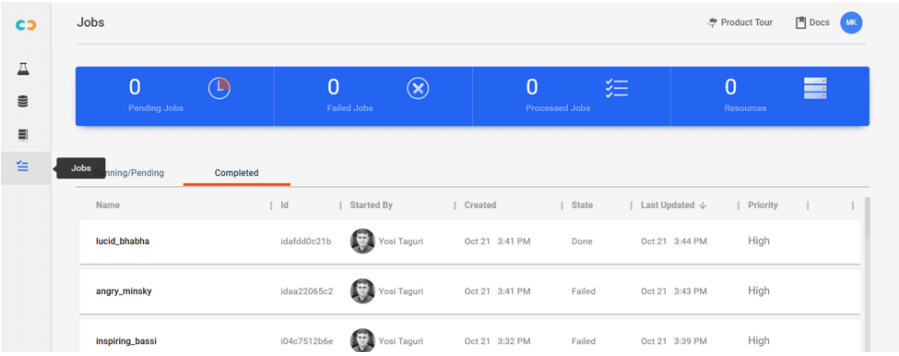
will need to distribute a ResNet model across multiple GPUs, and if performance is insufficient, scale out to multiple machines.

However, you'll find that running a deep learning model on multiple machines is difficult:

- **On-premises—**you will need to set up multiple machines for deep learning, manually run experiments and utilize resources.
- **In the cloud—**you can spin up machines quickly, but you will need to build and test machine images, and manually run experiments on each machine. You'll need to "babysit" your machines to ensure an experiment is always running, and to avoid wasting money with expensive GPU machines.

MissingLink solves all that. It's a deep learning platform that lets you scale out ResNet and other computer vision models automatically across numerous machines.



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To avoid idle time, MissingLink immediately runs another experiment when the previous one ends, and cleanly shuts down cloud machines when all jobs complete.

Learn more about the MissingLink platform.

Options for Running ResNet on TensorFlow

Using a Pre-Trained Model

The TensorFlow official models are a collection of example models that use TensorFlow's high-level APIs. The official TensorFlow Resnet model contains an implementation of ResNet for the ImageNet and the CIFAR-10 datasets written in TensorFlow.

You can download pre-trained versions of ResNet-50. There are four versions of ResNet-50 available with different precision accuracies. You can use transfer learning to speed up the process of training the pre-trained model. In transfer learning, a pre-trained model is used as a starting point of the next layer. In addition, you can freeze all of the layers, except the final fully connected layers, when fine-tuning your model.

Train TensorFlow ResNet From Scratch

While transfer learning is a powerful technique, you'll find it

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1. Launch TensorFlow environment with Docker

```
nvidia-docker run -it -v /data:/datasets
tensorflow/tensorflow:nightly-gpu bash
```

2. Download ImageNet

2.1 Clone the TPU repository

```
git clone https://github.com/tensorflow/tpu.git
```

2.2 install the GCS dependencies

```
pip install gcloud google-cloud-storage
```

2.3 Downloads the files from Image-Net.org

```
python imagenet_to_gcs.py --local_scratch_dir=/data/imagenet
--nogcs_upload
```

3. Download Official TensorFlow models

```
git clone https://github.com/tensorflow/models.git
```

4. Export PYTHONPATH

Export PYTONPATH to the models folder on your machine. Be sure to replace /datasets/models with your folder path.

```
export PYTHONPATH="$PYTHONPATH:/datasets/models"
```

5. Install Dependencies

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Scaling ResNet on TensorFlow with MissingLink

In this article, we learned the basics of ResNet and saw two ways to run TensorFlow ResNet:

- Using a pre-trained model and transfer learning
- Building ResNet components from scratch

Training ResNet is extremely computationally intensive, especially when working with a large number of layers. Don't wait for hours or days for ResNet to train. Use the MissingLink deep learning framework to:

- **Scale out ResNet** automatically across numerous machines, either on-premise or in the cloud.
- **Define a cluster of machines** and automatically run deep learning jobs, with optimal resource utilization.
- **Avoid idle time** by immediately running experiments one after the other, and shutting down cloud machines cleanly when the jobs are complete.

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Learn More About TensorFlow

7 Types of Neural Network Activation Functions: How to Choose?

A Recurrent Neural Network Glossary: Uses, Types, and Basic Structure

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